Terrain

The six terrains: exploration, battle fronts, unit multipliers, movement and what each is home to

Kingdoms Call reference sheet. Data file v2026.10.15-1202. Generated 2026-10-19 from the repository (branch at that date). Numbers are what the engine uses today, read straight from its tables; where a figure comes from the rulebook text instead, the section says so.

Provinces are plains, desert, forest, jungle, mountain or sea. Terrain decides how fast a province is charted, how wide a battle front is, which units thrive or suffer, which race's provinces lie there and which troops it offers. The per-unit multipliers are written once, in the engine's TERRAIN_MODS, and shown in full here.

What to look for when reviewing

- Is each terrain someone's home ground? Plains humans, forest elves, mountain dwarves, desert orcs; jungle belongs to nobody.
- How much the multipliers vary: cavalry ×0.70–0.75 in jungle and forest, dwarves ×1.10–1.15 in mountains, flyers ×0.75 under trees.
- Front width by terrain (R-089): mountains squeeze a battle to 2–4 units a side.
- Sea: a separate world with its own tax, lairs, guardians and penalties.

Where this comes from

- Engine: TERRAIN_MODS, KC_BASIC_EXPLORE, SIGNATURE_TERRAIN_P, battle/lines.js (front width), SEA_LAND_PENALTY, illusoryTerrainMult.
- Rulebook: The World & Provinces, Movement & Marching, The battle lines. Rulings R-081, R-089, R-100.

The six terrains

Terrain	Explore base (points per Explore at Explorer 0)	Battle front width	Signature race	Cavalry flanks	Who thrives	Who suffers	Other rules
Plains	35	6–8	Human	Yes (Tactical + 1 riders)	Humans ×1.10 line foot and longbows; horse ×1.15; Titans, Unicorns, Mammoths ×1.10	Dwarven Guard ×0.9, Mountain Guns ×0.85, Giants/Ents/Basilisks ×0.9	Ride Them Down (+6%/Free Lances to enemy losses)
Desert	25	6–8	Orcish	Yes	Horse ×1.15, Wargs ×1.10, Berserkers ×1.05, Basilisks and Titans ×1.10	Elven Bowmen ×0.8, Ents ×0.75, Bears ×0.8, Hydras and Mammoths ×0.85	Ride Them Down
Forest	20	4–6	Elven	No (Stag Riders and Unicorns still take a flank)	Elven units ×1.10–1.15 (and an Elven realm's own ×1.5); Ents ×1.15; Trolls, Bears, Unicorns, Berserkers ×1.10	Horse ×0.75, flyers ×0.75, Mammoths and Guns ×0.80, Titans ×0.8	Forest Wardens' Ambush (no enemy archery phase); Druidic Creatures ×1.5; Pagan Standing Stones +25% defending
Jungle	15	4–6	none	No (Stag Riders and Unicorns)	Elves ×1.10, Ents ×1.15, Basilisks and Hydras ×1.10	Horse ×0.70, flyers ×0.75, Guns and Titans and Mammoths ×0.75, line foot ×0.85	As forest
Mountain	20	2–4	Dwarven	No (Ironhoof Rams take a flank)	Dwarven Guard, Crossbows, Guns ×1.10, Dwarf Garrison ×1.15, Stone Giants and Trolls ×1.15, Bears ×1.10	Horse ×0.75, Wargs ×0.8, Stag Riders and Unicorns ×0.85, Titans ×0.8	Pagan Standing Stones +25% defending

Terrain	Explore base (points per Explore at Explorer 0)	Battle front width	Signature race	Cavalry flanks	Who thrives	Who suffers	Other rules
Sea	25	8–10 (no flanks)	none (wild sea-beasts and open water)	No	Fleets and sea-beasts ×1 everywhere; Kraken ×1.35 at sea; Shark Shoals swarm	Every land unit ×0.75 (Elven Galleys lift their archers); air-carried troops ×0.25, their carriers ×0.5	Only fleets, carried troops, flyers or Part Sea may enter. Tax 1 open water, 2–3 whale and shark waters, 4–6 the sea-monsters' waters. Lairs and guardians of its own.

Unit strength multipliers by terrain (the engine's table in full)

Fleets and sea-beasts read ×1 everywhere. Flyers read ×1 except ×0.75 in forest and jungle. A unit not listed in a terrain reads ×1.

Unit	Family	Type	Plains	Desert	Forest	Jungle	Mountain	At sea	Under Illusory Terrain
Free Lances	Human	cavalry	×1.15	×1.15	×0.75	×0.70	×0.75	×0.75 (sea penalty)	×0.70
Galleon Fleet	Human	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Gilded Lances	Human	cavalry	×1.15	×1.15	×0.75	×0.70	×0.75	×0.75 (sea penalty)	×0.70
Human Garrison	Human	infantry	×1	×0.90	×1	×1	×1	×0.75 (sea penalty)	×0.85
Men-at-Arms	Human	infantry	×1.10	×1	×0.90	×0.85	×0.90	×0.75 (sea penalty)	×0.85
Noble Griffons	Human	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Shieldwall Levies	Human	infantry	×1.10	×1	×0.90	×0.85	×0.90	×0.75 (sea penalty)	×0.85
Yeoman Longbows	Human	ranged	×1.10	×0.90	×0.85	×0.80	×1	×0.75 (sea penalty)	×0.80
Eagle Riders	Elven	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Elven Bowmen	Elven	ranged	×1	×0.80	×1.15	×1.10	×1	×0.75 (sea penalty)	×0.80
Elven Galleys	Elven	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Elven Garrison	Elven	infantry	×1	×0.85	×1.10	×1.10	×0.90	×0.75 (sea penalty)	×0.85
Forest Wardens	Elven	infantry	×1	×0.90	×1.10	×1.10	×0.85	×0.75 (sea penalty)	×0.85
Shadow Archers	Elven	ranged	×0.90	×0.85	×1.10	×1.10	×1	×0.75 (sea penalty)	×0.85
Stag Riders	Elven	cavalry	×1.10	×1	×1.10	×1	×0.85	×0.75 (sea penalty)	×0.85
Dwarf Garrison	Dwarven	infantry	×0.90	×0.85	×0.90	×0.85	×1.15	×0.75 (sea penalty)	×0.85
Dwarf Ironclads	Dwarven	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Dwarven Guard	Dwarven	infantry	×0.90	×0.90	×0.90	×0.85	×1.10	×0.75 (sea penalty)	×0.85
Gnomish Gyrocopters	Dwarven	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Ironhoof Rams	Dwarven	cavalry	×1	×0.90	×0.90	×0.85	×1	×0.75 (sea penalty)	×0.85
Ironwood Crossbows	Dwarven	ranged	×1	×1	×0.90	×0.85	×1.10	×0.75 (sea penalty)	×0.85
Mountain Guns	Dwarven	ranged	×0.85	×0.85	×0.80	×0.75	×1.10	×0.75 (sea penalty)	×0.75
Black Arrows	Orcish	ranged	×1	×1	×0.90	×0.90	×0.90	×0.75 (sea penalty)	×0.85
Goblin Horde	Orcish	infantry	×1	×1	×1	×1	×0.90	×0.75 (sea penalty)	×0.85
Orc Berserkers	Orcish	infantry	×1	×1.05	×1.10	×1.05	×0.90	×0.75 (sea penalty)	×0.85
Orc Garrison	Orcish	infantry	×1	×0.90	×1	×1	×0.90	×0.75 (sea penalty)	×0.85
Orcish Longships	Orcish	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—

Unit	Family	Type	Plains	Desert	Forest	Jungle	Mountain	At sea	Under Illusory Terrain
Warg Riders	Orcish	cavalry	×1.10	×1.10	×0.90	×0.80	×0.80	×0.75 (sea penalty)	×0.80
Wyvern Riders	Orcish	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Crimson Rocs	Creature	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Forest Ents	Creature	infantry	×0.90	×0.75	×1.15	×1.15	×0.90	×0.75 (sea penalty)	×0.75
Great Bears	Creature	infantry	×1	×0.80	×1.10	×1	×1.10	×0.75 (sea penalty)	×0.80
Great Turtles	Creature	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Mammoth Host	Creature	infantry	×1.10	×0.85	×0.80	×0.75	×1	×0.75 (sea penalty)	×0.75
Stone Giants	Creature	infantry	×0.90	×0.90	×0.90	×0.80	×1.15	×0.75 (sea penalty)	×0.80
Thunderbirds	Creature	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Unicorn Herd	Creature	cavalry	×1.10	×1	×1.10	×0.90	×0.85	×0.75 (sea penalty)	×0.85
Whale Cohort	Creature	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Basilisk Brood	Monster	infantry	×0.90	×1.10	×1	×1.10	×0.90	×0.75 (sea penalty)	×0.85
Cave Trolls	Monster	infantry	×1	×0.85	×1.10	×1.05	×1.15	×0.75 (sea penalty)	×0.85
Dread Hydras	Monster	infantry	×1	×0.85	×1.05	×1.10	×0.90	×0.75 (sea penalty)	×0.85
Elder Dragons	Monster	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Eldritch Kraken	Monster	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Manticore Pride	Monster	flyer	×1	×1	×0.75	×0.75	×1	×0.75	×0.75
Sea Giants	Monster	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Shark Shoals	Monster	naval	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×1 (naval)	×0.75	—
Titan Warbeasts	Monster	infantry	×1.10	×1.10	×0.80	×0.75	×0.80	×0.75 (sea penalty)	×0.75

At sea: every land unit that does not fly fights at ×0.75 (the sea penalty, applied on top of this table). Illusory Terrain: a non-caster force fights as if in the worst land terrain for each unit, always at least ×0.85.

Terrain and the wider rules

Rule	What it is
Exploration	Each Explore charts (base) × (Explorer level + 1) points, Explorer counted to 5. A plains province goes 35 to 100% in three orders at Explorer 0.
Front width	Rolled per battle, same for both sides: plains and desert 6–8, forest and jungle 4–6, mountain 2–4, sea 8–10. The side that knows the ground better may take the roll or one wider or narrower. Behind the front: second line ×0.5 (ranged over a fighter ×1), reserve ×0.25, beyond ×0.125.
Flanks	Tactical + 1 cavalry ride the flanks on plains and desert only. Stag Riders and Unicorns take a flank in forest and jungle; Ironhoof Rams in the mountains.
Movement	Every land step costs 1 point whatever the terrain; sea steps cost a sea point. Terrain never slows a march; unit speed does.
Race placement	Each race is about six times as likely to be placed in its signature terrain; about two thirds of a terrain's provinces carry its race. Jungle has no signature race.
Wild country	Creature and Monster provinces hold one beast type each and a ritual site; Creatures tend to forest, mountain and plain, Monsters to mountain, desert and jungle, by their terrain tables.
World shape	40% sea in coherent coasts and inland seas; the land shared equally between the four races, with a portion of wild country. Map size by realm count: 2 realms 8×5 to 9–10 realms 16×11.
Homeward Flight	Elven units of an Elven realm escape death at 30% in desert or mountain, 45% on plains, 60% in forest or jungle.
Standing Stones	The Pagan Wonder: armies fight +25% defending forest, jungle or mountain.

Lair site names by terrain (the older terrain-named pools)

Terrain	Site names
Plains	Ancient Battlefield, Ruined Watchtower, Goblin Camp, Roadside Shrine, Bandit Stronghold, Old Graveyard, Collapsed Aqueduct, Abandoned Fort, Plague Village, Sunken Road Shrine, Mercenary Encampment, Burning Mill
Desert	Buried Pyramid, Sand Drake Lair, Oasis Temple, Forgotten Tomb, Sphinx Lair, Glass City, Scorpion Den, Lost Caravan Ruins, Sand Dune Fortress, Sun God Shrine, Bone Maze, Mirage Palace
Forest	Haunted Grove, Elven Ruins, Spider Warren, Druid Circle, Dark Elf Camp, Wolf Den, Witch's Hut, Overgrown Barrow, Treant Hollow, Bandit Cave, Ruined Monastery, Fey Crossing
Jungle	Lost City, Serpent Temple, Giant Insect Nest, Overgrown Fortress, Sunken Palace, Tribe Village, Vine-Choked Ziggurat, Stone Idol, Piranha Pool, Cursed Well, Monkey King Lair, Poison Garden
Mountain	Dragon Eyrie, Dwarven Mine, Troll Bridge, Giant's Hall, Frozen Crypt, Eagle Aerie, Crystal Cave, Old Observatory, Collapsed Mine Shaft, Golem Workshop, High Temple, Glacier Tomb
Sea	Sea Cave, Pirate Cove, Sunken Ship, Kraken Lair, Mermaid Grotto, Ghost Galleon, Coral Fortress, Lighthouse Ruin

World generation now names a lair from its guardian's category pool (see the Lairs sheet). These terrain pools are still read by generate.js in two places; check whether they are still wanted.

Gaps, questions and things to check

These are observations from reading the data files, the engine and the rulebook together. None of them is a ruling. Each is a question for Toby or a place where two sources disagree.

#	Observation
1	Jungle has no signature race and is the harshest terrain for almost everyone (cavalry ×0.70, line foot ×0.85, guns ×0.75); only elves, ents, basilisks and hydras like it. Is it meant to be the empty quarter?
2	Mountain fronts are 2–4 units wide, so a mountain battle is decided by the two to four strongest units a side; numbers barely count. That rewards elite dwarves and giants heavily. Confirm it is wanted.
3	Elven units carry both a per-unit forest bonus (×1.10–1.15) and the realm-wide Woodland Mastery ×1.5, so an Elven realm's bowmen in a forest fight at about ×1.7 before kinship; with a king of their own blood about ×2.5 (before Tactical). Check against the dwarves' ×1.5 defending plus ×1.1–1.15 in the mountains.
4	Desert is the orcs' home but their units get only ×1.05–1.10 there; the human horse gets ×1.15 in the desert too. The orcish home-ground edge is small.
5	The sea penalty ×0.75 applies to every non-flying land unit, and nothing removes it for infantry (Naval 8 aside). Landings are therefore mostly a fleet-strength fight.
6	Sea tax (1 for open water) is the one place a province pays less than 2 (R-015 open question).
7	Terrain does not change movement cost (R-081), only exploration and battle; the rulebook never says a mountain march costs the same as a plains march. A line would help.
8	Two sets of lair names exist: the guardian-category pools (used for naming) and the older terrain pools (still referenced by generate.js). One set should go or their roles be stated.